Reading the Atmosphere Profile

The Skew-T diagram of MODE-S Wind (Live Map and Sounding page)

Otso Laakso (OH2GAX)
Tuusula, Finland · near Helsinki-Vantaa airport (EFHK)

In short. The Atmosphere Profile panel on the right of the Live Map is a small *Skew-T log-p* diagram: a vertical profile of temperature and wind, like a weather balloon sounding. Its data do not come from a balloon but from the aircraft themselves. Their transponders report airspeed, Mach number, heading and track, and from these MODE-S Wind computes the air temperature and the wind around each aircraft. The grey curve is the average of all aircraft within 150 km over the last 60 minutes; clicking an aircraft adds its own profile in colour. An Analysis button marks the tropopause, the freezing level, jet streams, inversions, stability, turbulence risk, warm and cold advection, contrail levels, the daytime thermal top and — with the METAR dewpoint — the cloud base of a surface air parcel, its convective energy and stability index. The Airport selector adds a fine low-level profile (200 ft steps) from the landings and departures at Helsinki-Vantaa. The Sounding page uses the same diagram for past hours and single flights. Aircraft report no humidity, so the analysis knows moisture only at the ground, from the METAR; the Jokioinen radiosonde and the MEPS weather model can be drawn beside the profile for comparison, with their humidity aloft.
New in version 4: comparison profiles beside the aircraft — the radiosonde (Sonde), the MEPS weather model (MEPS) and the trial moisture strip (RH), and why the model has fewer levels than the aircraft profile (Section 12).
Version 3 added: the data-quality indicator — a badge with the number of aircraft behind the profile, rings on layers measured by only one or two aircraft, and the Data quality tick box (Section 5); why the range bars are wider low down (Section 3).
Version 2 added: airport low-level profile (Section 9); METAR dewpoint and surface parcel — LCL, CAPE, Lifted Index, fog / low stratus flag (Section 10.5); the Sounding page with the same diagram (Section 11); per-airframe heading correction of the winds.

Contents

1 What the panel shows and where **2**

2 Upper-air parameters in brief **2**

3 Where the data come from **3**

4 The diagram, element by element **4**

5 How much data: the data-quality indicator **5**

6 Reading the profile **6**

7 A selected aircraft **7**

8 Smoothing **7**

9 The airport low-level profile **7**

10 The Analysis button **8**

10.1 Levels . . . 9
10.2 Stability . . . 9
10.3 Turbulence . . . 10
10.4 Advection and contrails . . . 10
10.5 Full: thermal top and surface parcel . . . 11

11 The Sounding page **12**

12 Comparison profiles: radiosonde, MEPS model and RH strip **13**

13 Limitations **15**

14 Quick reference **16**

A Formulas **16**

1 What the panel shows and where

The panel fills the right-hand side of the Live Map page (Fig. 1). It always shows the **area profile**: all aircraft within 150 km of the receiver over the last 60 minutes, refreshed every 5 minutes. When you click an aircraft on the map or in the list, its own profile is drawn on top in colour; click an empty spot on the map, press Esc or close the aircraft's popup to return to the area profile. The Sounding page draws exactly the same diagram, larger, for the area profile of any window, a stored past hour or a single flight (Section 11).

Under the diagram are three controls: Density (how many points of a selected aircraft are drawn), Smooth (vertical smoothing of the area temperature profile) and Airport (the airport low-level profile, Section 9). In the panel title are the Data quality tick box, which shows how many aircraft the profile rests on (Section 5), and the Analysis button, which steps through six levels of automatic analysis (Section 10). Hovering over the diagram shows the values of the nearest level.

A Skew-T is read from bottom to top. Height is not drawn in metres but as *air pressure*, which falls with height; temperature lines are tilted ("skewed") to the right. This looks unusual at first, but it makes the important features of the atmosphere stand out: how fast temperature falls with height, where it suddenly stops falling, and where the air is stable or unstable.

2 Upper-air parameters in brief

Pressure (hPa). The weight of the air above. About 1013 hPa at sea level, 850 hPa at about 1.5 km, 700 hPa at 3 km, 500 hPa at 5.5 km, 300 hPa at 9 km and 250 hPa at 10.4 km. Meteorologists use pressure as the vertical coordinate because weather maps and soundings are made at fixed pressure levels.

Pressure altitude and flight level (FL). The altitude an aircraft reports is its *pressure altitude*: the height in the International Standard Atmosphere (ISA) at which the measured pressure would occur, always referenced to 1013.25 hPa. FL350 means 35 000 ft pressure altitude. Pressure altitude is therefore a direct measure of pressure, not of true height above the sea.

True altitude. The real height. It differs from pressure altitude by about 27 ft for each hPa that the sea-level pressure (QNH) differs from 1013, and further in air warmer or colder than ISA (colder air is denser, so the same pressure lies lower).

Temperature and ISA. The ISA reference falls by 6.5 °C per km (about 2 °C per 1 000 ft) from +15 °C at sea level to −56.5 °C at 11 km and stays constant above. Real air is warmer or colder; the difference is the *ISA deviation*.

Lapse rate. How fast temperature falls with height, in K/km. 6.5 K/km is the ISA average; 9.8 K/km is the *dry adiabatic* rate at which unsaturated rising air cools; 4–7 K/km is the *moist adiabatic* rate of saturated air (slower, because condensing water releases heat).

Stability. If the air's lapse rate is smaller than that of a rising air parcel, the parcel becomes colder than its surroundings and sinks back: the layer is *stable*. If larger, the parcel keeps rising: *unstable*. Between the dry and moist rates the layer is *conditionally unstable*: stable for clear air, unstable once the air is saturated (showers, cumulus).

Inversion. A layer where temperature *rises* with height. Very stable; it traps haze, fog and low cloud below it, and often forms near the ground on clear nights.

Freezing level. The height of the 0 °C isotherm. With a warm layer aloft there can be more than one.

Tropopause. The top of the weather layer (troposphere), where temperature stops falling with height; typically 8–12 km over Finland, lower in cold air and lows, higher in warm air.

Wind. Direction the wind blows *from* (degrees true) and speed in knots. The strongest winds, the *jet stream*, are usually just below the tropopause.

Wind shear. Change of wind with height. Strong shear in stable air can still overturn and cause clear-air turbulence.

Advection. Transport of warmer or colder air by the wind. It changes the profile over the following hours.

Humidity, dewpoint. The amount of water vapour. The *dewpoint* is the temperature to which the air must be cooled to become saturated; the *spread* (temperature minus dewpoint) is zero in fog and cloud. A radiosonde measures humidity all the way up; aircraft transponders do not report it. MODE-S Wind therefore knows the dewpoint **only at the ground**,

from the METAR (Section 13).

Lifting condensation level (LCL). The height at which air lifted from the ground cools (along the dry adiabat) to its dewpoint and condenses: the base of cumulus clouds formed by surface heating. Roughly 400 ft per degree of spread.

CAPE, CIN, Lifted Index. *CAPE* (convective available potential energy, J/kg) is the energy gained by a lifted parcel while it is warmer than its surroundings — a measure of how strong showers and thunderstorms can become (below about 100 weak, above 1 000 considerable). *CIN* is the energy needed to lift it to the level where it becomes buoyant. The *Lifted Index* is the environment minus parcel temperature at 500 hPa: negative means unstable.

3 Where the data come from

MODE-S Wind listens to the 1090 MHz replies of aircraft transponders. Many airliners answer ground radar interrogations with *Enhanced Surveillance* (EHS) reports:

- **BDS 5,0** (track and turn): true track, groundspeed, true airspeed (TAS), roll angle;
- **BDS 6,0** (heading and speed): magnetic heading, Mach number, indicated airspeed, vertical rate;
- **BDS 4,4** (meteorological routine air report, MRAR): wind and temperature measured on board — reported by only a few aircraft;
- and the aircraft's **ADS-B** broadcasts with position, altitude and velocity.

Temperature from Mach and TAS. The air-data computer measures both the true airspeed and the Mach number. Their ratio is the speed of sound, which depends only on the air temperature:

$$T\,[\mathrm{K}] = 288.15 \left(\frac{\mathrm{TAS}}{661.47\,\mathrm{kt} \times M} \right)^2 .$$

A single reply pair scatters by about ± 1.3 K at cruise and ± 3–4 K on approach, because Mach is transmitted in steps of 0.004. Layer averages of many replies are much better: compared with the Jokioinen radiosonde (8 October 2026, 12 UTC) the mean difference was $+0.7$ K with a standard deviation of 1.6 K. The rare MRAR temperature is used first when available.

Why the range bars are wider low down. Mach is sent in steps of 0.004 and the TAS in steps of 2 kt. At cruise (Mach 0.78, 450 kt) one step changes the computed temperature by about 2 K; at approach and climb speeds (Mach 0.25, 160 kt) by roughly 7–9 K. Single replies near the ground therefore scatter widely, and the 10–90 % bars of the low layers are wide even when the aircraft agree. Typical daytime bar widths at EFHK (28 September – 11 October 2026): about 13–14 K at 0–1 000 ft, 12 K at 1 000–2 000 ft, 8–9 K at 2 000–4 000 ft and 5–6 K from 6 000 ft up; turbulent days widen them further. The layer mean averages dozens of replies and is much steadier than the bars suggest: read the line, and use the bars as a measure of the scatter. More replies make the mean steadier but do not narrow the bars.

Wind from the wind triangle. The aircraft moves over the ground with its *ground vector* (track and groundspeed) and through the air with its *air vector* (true heading and TAS). The difference is the wind:

$$\vec{V}_{\mathrm{wind}} = \vec{V}_{\mathrm{ground}} - \vec{V}_{\mathrm{air}}.$$

The heading is corrected for magnetic declination and, since 10 October 2026, for each airframe's own heading offset, which the system learns by comparing the aircraft's winds with those of the other aircraft at the same level (on average 1.5°, which would otherwise mean about 8 kt of wind error at cruise). Replies taken during turns are rejected. MRAR winds are preferred when reported.

Pressure. Each observation's pressure is the ISA pressure of its pressure altitude — which, by the definition of pressure altitude, is the actually measured static pressure.

The area profile. Observations from the last 60 minutes within 150 km of the receiver are grouped into layers 1 000 ft thick up to 10 000 ft and 2 000 ft thick above. For each layer:

- **temperature** = 10 % trimmed mean (the warmest and coldest 10 % removed), with the 10–90 % range and the number of replies;
- **wind** = vector mean of all winds in the layer, with its scatter (SD) and number.

A typical daytime hour gives 15–25 layers from 25–70 different aircraft, about 20–35 per layer; around midnight UTC only 2–5 aircraft fly in the area, often a single one per layer. The data-quality indicator shows this (Section 5).

The airport low-level profile. Every landing and departure at EFHK is followed from 3 000 ft down to 200 ft (or up), and *every* raw BDS 5,0 / 6,0 reply — not only the stored observations — is turned into a temperature and, with wings level, a wind for its 200 ft band. The bands of all flights of the shown period are pooled (Section 9).

Surface values from the METAR. The airport's METAR (every 30 minutes, fetched every 10) gives the QNH, the air temperature, the dewpoint and the cloud groups. They are used for the ground line, the surface parcel of the analysis, and are stored with each landing and each hourly profile.

4 The diagram, element by element

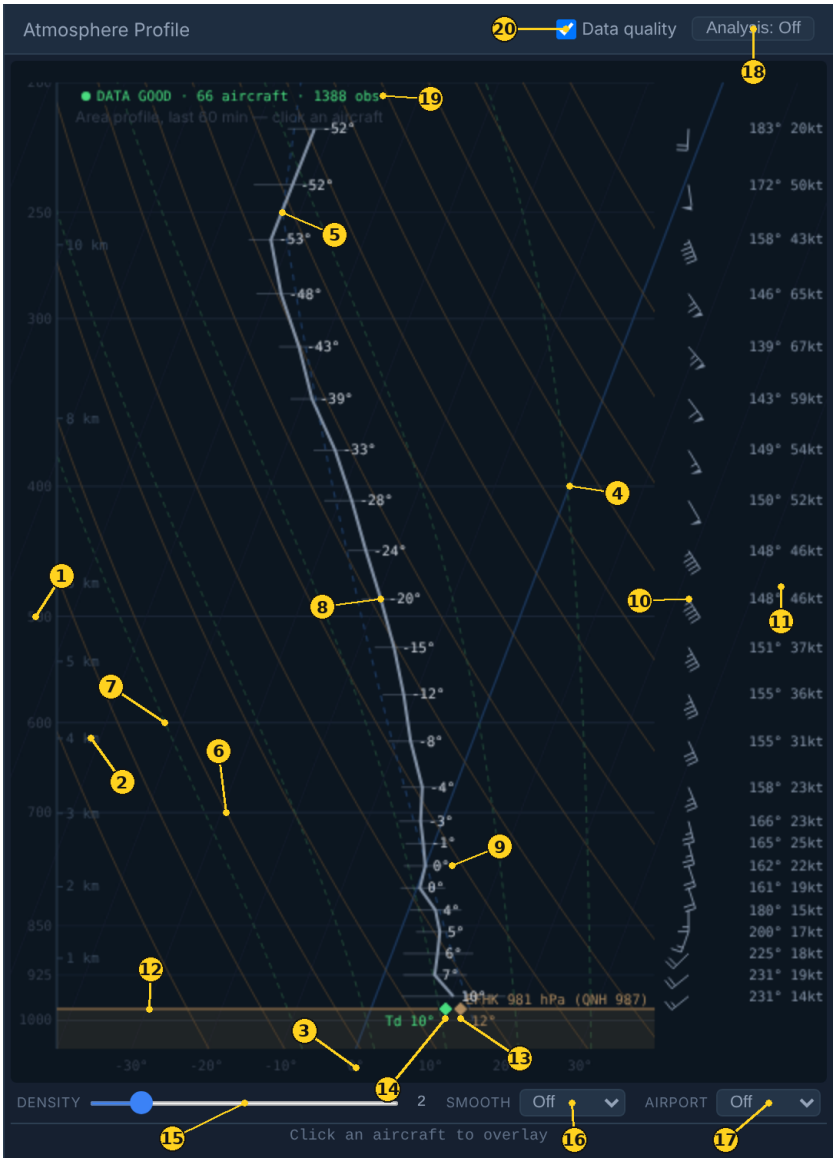
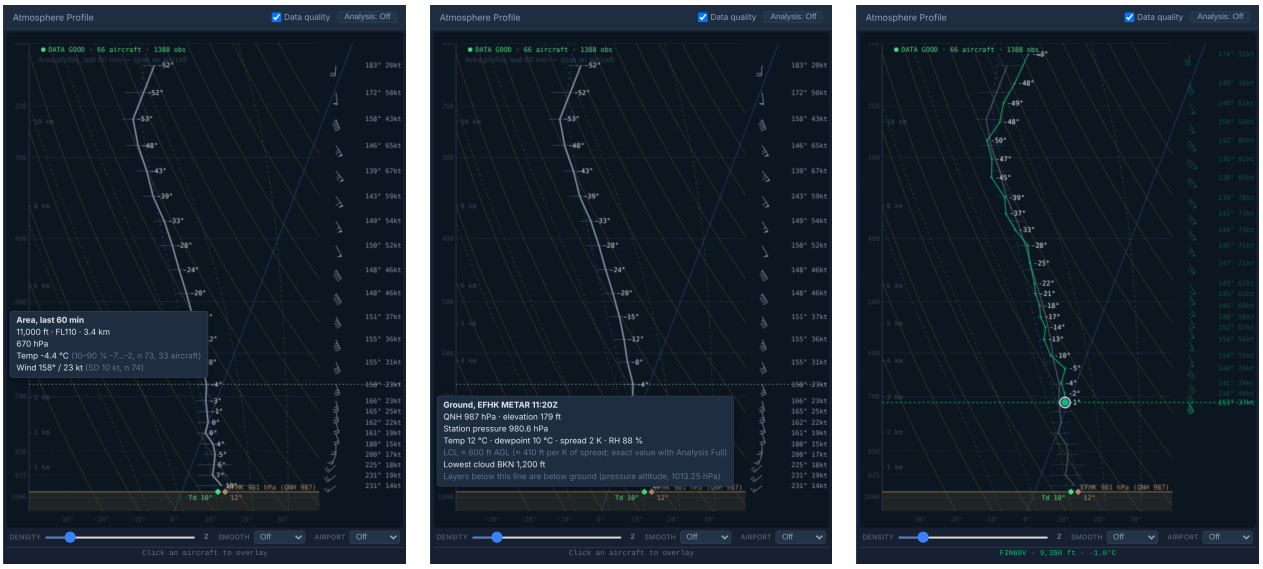


Figure 1. The Atmosphere Profile panel with the area profile of 9 October 2026, 11–12 UTC (Analysis off; 66 aircraft, 1 388 observations). The numbers refer to the list in Section 4. The surface point uses an illustrative METAR (QNH 987 hPa, 12 °C, dewpoint 10 °C, BKN012).

(1) Pressure axis (hPa), logarithmic: equal distances mean equal pressure *ratios*, which is close to equal height steps.
(2) Height markers (1–12 km), pressure altitude. Use them for orientation; the true height differs a little (Section 2).
(3) Temperature scale (−30 … +30 °C), valid along the bottom of the plot. Higher up you must follow the tilted isotherms.
(4) Isotherms: lines of equal temperature, tilted 45°-like to the right; the 0 °C isotherm is drawn stronger (blue). A temperature is read by following its isotherm, not by looking straight down.
(5) ISA reference (dashed blue): the standard-atmosphere temperature. Curve left of it = colder than ISA, right = warmer. On this day the measured profile is almost on top of it.
(6) Dry adiabats (faint solid brown, every 10 K): the path of rising or sinking unsaturated air.
(7) Moist adiabats (faint dashed green): the path of rising saturated air. They are steeper than the dry adiabats near the ground and become parallel to them high up, where the air holds almost no water.
(8) Area temperature profile (grey line) with the layer values in °C.
(9) 10–90 % range of each layer (thin horizontal bar): a wide bar means the replies disagreed, because of few replies, low speed (approach and climb, Section 3) or real differences across the 150 km area.
(10) Wind barbs: the staff points into the wind direction; a half barb is 5 kt, a full barb 10 kt, a pennant 50 kt.
(11) Wind values: direction (° from) and speed (kt) of each layer.

(12) **Ground line** (brown, with shading below): the station pressure of EFHK from the current METAR QNH and the airport elevation (about QNH − 6.5 hPa). The label reads e.g. `EFHK 981 hPa (QNH 987)`. Everything below the line is underground: with a low QNH the lowest pressure-altitude layer may lie below the surface and is then empty.
(13) **Surface temperature** (brown diamond) from the METAR.
(14) **Surface dewpoint** (green diamond, `Td 10°`) from the METAR. The horizontal distance between the two diamonds is the spread: they touch in fog or with a very low cloud base.
(15) Density slider: thins the points drawn for a selected aircraft (Section 7).
(16) Smooth selector: vertical smoothing of the area profile (Section 8).
(17) Airport selector: Off / Overlay / Zoom, the airport low-level profile (Section 9).
(18) Analysis button: automatic analysis, six levels (Section 10).
(19) **Data badge**: how much data the profile rests on — `DATA GOOD · 66 aircraft · 1388 obs` (Section 5).
(20) Data quality tick box: shows or hides the badge and the rings of single-aircraft layers (Section 5).

Hover read-out. Moving the mouse over the diagram shows the nearest level (Fig. 2a): its pressure altitude in ft, as a flight level and in km, the pressure, the temperature with its 10–90 % range, number of replies and number of aircraft, and the wind with its scatter (SD) and number. A wide range or a small number tells you how far to trust the value; a layer from only one or two aircraft gets a red warning line (Section 5). Hovering the ground line shows the METAR time, QNH, station pressure, temperature, dewpoint, spread, relative humidity, a quick cloud-base estimate (about 410 ft per degree of spread) and the lowest METAR cloud group (Fig. 2b).



(a) Hover read-out of the 11 000 ft layer: 670 hPa, −4.4 °C (10–90 % −7…−2 °C, 73 replies, 33 aircraft), wind 158°/23 kt (SD 10 kt, 74 winds).

(b) Hover on the ground line: METAR temperature 12 °C, dewpoint 10 °C, spread 2 K, RH 88 %, LCL ≈ 800 ft, lowest cloud BKN 1 200 ft.

(c) FIN69V descending from FL390 towards EFHK (green) over the grey area profile; the circle is its present position (9 350 ft, −1 °C) with a dotted line to its current wind.

Figure 2. Hover read-outs and a selected aircraft (9 October 2026, 11–12 UTC).

5 How much data: the data-quality indicator

The area profile is only as good as the traffic behind it. By day dozens of aircraft cross the area every hour; late at night only a handful, and a layer may then rest on a single aircraft — one set of instruments, with no other aircraft to average out its errors. The **data badge** at the top left of the diagram says how much the profile rests on (Fig. 3):

`● DATA GOOD · 66 aircraft · 1388 obs`

the number of *different* aircraft and of observations in the window, graded by aircraft per hour of window:

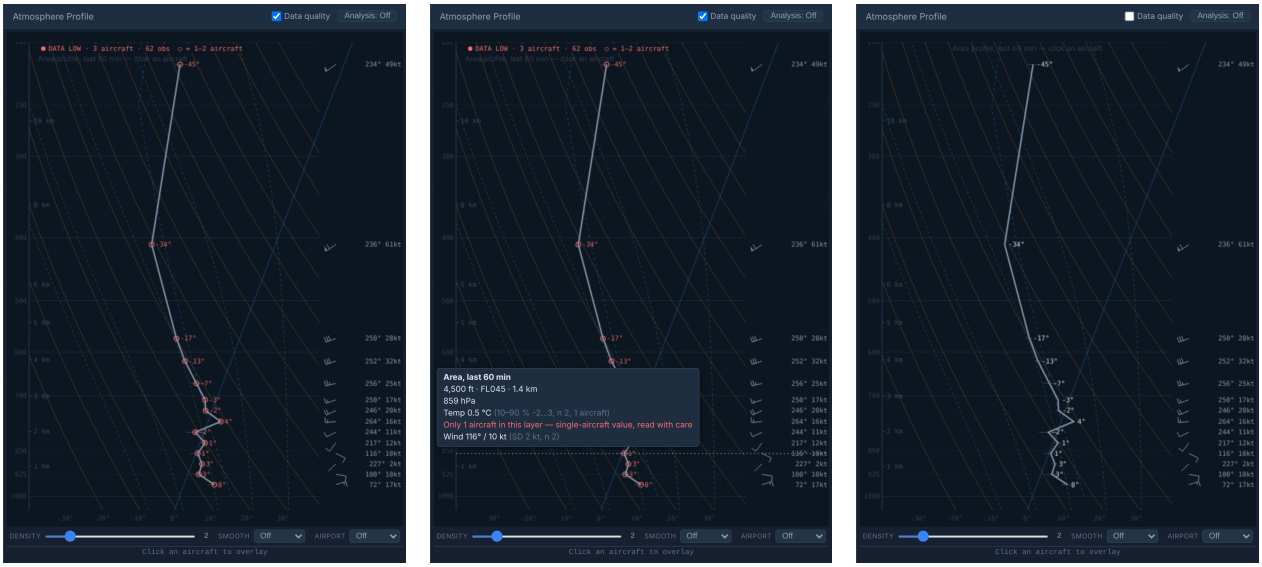
- **GOOD** (green): 20 or more — typical from about 03 to 21 UTC;
- **FAIR** (amber): 8–19 — usually the evening and early-morning hours (22 and 02 UTC);
- **LOW** (red): fewer than 8 — every night around 23–00 UTC and on most nights at 01 UTC (02–04 Finnish summer time), with 1–5 aircraft.

Rings: single-aircraft layers. A layer measured by only one or two aircraft gets a small **red ring** on its temperature point and a red value, and the badge adds the key `○ = 1–2 aircraft`. Its value can be off by several degrees or knots:

one aircraft's temperature scatters by 1–4 K and its wind carries that aircraft's own heading error. By day the rings usually appear only at the very top (above FL380, where few airliners cruise; Fig. 8b) and occasionally in the lowest layer; at night on almost every layer. The hover read-out of such a layer gives the number of aircraft and the warning `Only 1 aircraft in this layer -- single-aircraft value, read with care` (Fig. 3b).

What to do with a LOW profile. Read it as a rough guide. The automatic analysis still runs on such layers, so inversions, shear or turbulence flagged in a LOW profile deserve extra caution — a single aircraft's step in temperature or wind looks like a real feature. A selected aircraft's own profile is not graded: it always comes from one aircraft.

The tick box. Data quality in the panel title (ticked by default) shows or hides the badge, the rings and the hover warning; the aircraft counts stay in the hover read-out. The setting is remembered in the browser and shared with the Sounding page, where the badge works the same way for Area · now and for stored hours that carry the aircraft count, and the table has an A/C column with the number of aircraft per layer (Section 11).



(a) 11 October 2026, 00–01 UTC: `DATA LOW · 3 aircraft · 62 obs`; every layer is ringed.

(b) Hover on the 4 500 ft layer: 2 replies from 1 aircraft, with the warning.

(c) The same with Data quality unticked.

Figure 3. A night profile from three aircraft. Compare with the daytime profile of Fig. 1 (66 aircraft). No METAR is drawn here.

6 Reading the profile

1. **Start at the ground line** and the surface temperature. Compare the lowest layers with it: if the air just above is *warmer* than the surface, there is a surface inversion (typical on clear nights and mornings). Look at the dewpoint diamond: a spread of 0–2 K means fog, mist or a low cloud base is likely; a large spread means dry air and a high cloud base.
2. **Follow the curve upwards** and compare its slope with the adiabats:
   - steeper (leaning more to the left) than the dry adiabats: *unstable* — rare except right above sun-heated ground;
   - between the dry and the moist adiabats: *conditionally unstable* — showers and cumulus are possible if the air is moist enough;
   - less steep than the moist adiabats: *stable*;
   - vertical along an isotherm or leaning right: *isothermal layer or inversion* — very stable, a lid for haze, fog and low cloud.
3. **Find the freezing level** where the curve crosses the bold 0 °C isotherm, and the −10 and −20 °C isotherms. Between 0 and −20 °C icing is possible *if* the aircraft is in cloud.
4. **Find the tropopause**: the bend where the curve turns from sloping to vertical or even to the right. Above it, in the stratosphere, temperature stays nearly constant.
5. **Read the winds**: speed usually increases towards the tropopause (jet stream) and decreases above it. Wind *turning* with height tells about advection: clockwise (veering) means warm air arriving, anticlockwise (backing) cold air.
6. **Compare with ISA**: a profile well left of the dashed blue line is a cold air mass, right of it a warm one. Aircraft performance (TAS, climb rate) depends on this.

Example (Fig. 1). On 9 October 2026 at 11–12 UTC a deep low (QNH 987 hPa) lay over southern Finland. The profile stays close to ISA all the way up. Just above the ground the temperature falls quickly (10 °C at 1 500 ft, 7 °C at 2 500 ft); between 2 500 and 4 500 ft it falls only slowly, a stable layer; at 6 500–7 500 ft it stays near 0 °C, so the freezing level

(FL073) lies in an almost isothermal layer. Above, the curve follows the moist adiabats fairly closely, as is typical of a saturated low. It bends at about 260 hPa: the tropopause is at FL330 and −52 °C, low because of the low. A south-easterly jet of 67 kt blows at FL290. The winds *back* from 230° near the ground to 160° at 6 500 ft — cold air arriving at low levels (Fig. 7b).

7 A selected aircraft

Clicking an aircraft draws its own profile in its map colour (Fig. 2c). It combines the observations stored for the current flight (loaded when you select it) with new live data, so a descending arrival builds a complete profile from cruise to the runway. The circle marks the latest point; its values are shown in the line under the diagram. Wind barbs and values on the right then belong to the aircraft, and the area profile stays in the background in grey.

A single aircraft is noisier than the area average, especially at low speed: compare it with the grey curve rather than reading single points. The Density slider (1–8) draws a point only every 400× density ft of altitude change, to keep the labels readable. The aircraft's track on the map can be shown with the Track checkbox.

8 Smoothing

The Smooth selector applies a vertical running mean to the temperatures of the area profile: 3 layers (weights 1-2-1) or 5 layers (1-2-3-2-1), each layer also weighted by the square root of its number of replies. Smoothing removes layer-to-layer jitter and makes it easier to compare the slope of the curve with the adiabats (Fig. 4). It is for display only: the hover read-out then shows both the smoothed value and the layer's own value, and the 10–90 % bars stay unsmoothed. Strong smoothing also softens real features — inversions and the tropopause — so switch it off when you look at those. The automatic analysis always uses its own light smoothing, whatever this selector shows.

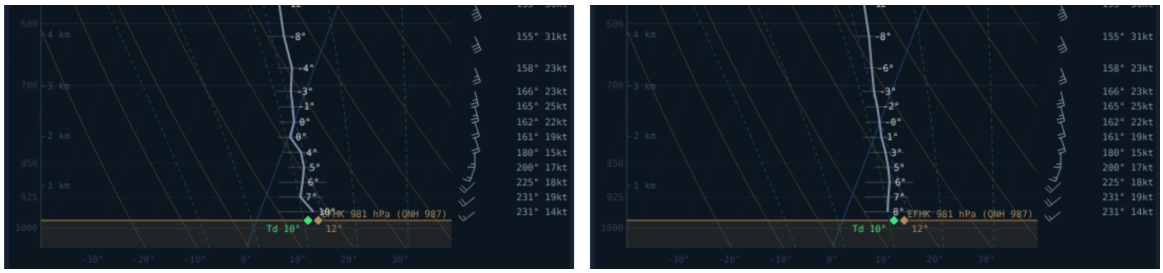


Figure 4. The lowest 4 km of the profile of Fig. 1 without smoothing (left) and with 5 layers (right).

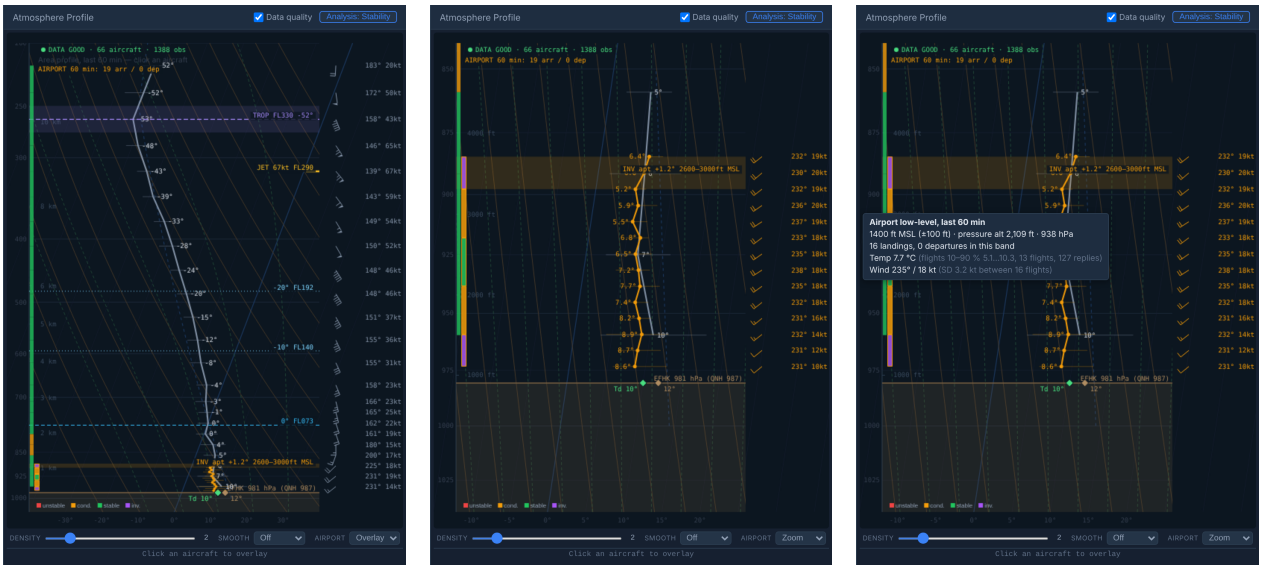
9 The airport low-level profile

The area profile has 1 000 ft layers and averages over 150 km. Close to the ground, where inversions, fog and low-level wind shear matter most, that is coarse. The Airport selector adds a second profile in amber, made only from the aircraft landing and departing at Helsinki-Vantaa:

- **200 ft bands** from 200 to 3 000 ft *above mean sea level* (MSL, from the METAR QNH), labelled in ft MSL;
- from **every raw reply** on final approach or in the climb-out, so a busy hour gives some hundred replies per band;
- per band: the reply-weighted temperature (dots, with the values), the spread between flights (10–90 %, thin bars), and the vector-mean wind with the barbs and values at the right;
- the period is the same as the area profile: the last 60 minutes on the Live Map, the selected window, hour or flight (±15 min) on the Sounding page.

The legend at the top left gives the number of landings and departures used, e.g. `AIRPORT 60 min: 19 arr / 0 dep`.

Overlay draws it on the normal diagram, where it occupies only the bottom 1 000–900 hPa (Fig. 5a). **Zoom** changes the diagram to the surface … about 840 hPa (5 000 ft), with less skew, finer isobars and the airport winds at the right (Fig. 5b); the area profile stays in grey for comparison. With Analysis at Stability or higher, significant inversions of the airport profile are shaded in amber (`INV apt +1.2° 2600–3000ft MSL`) and a second stability strip with an amber frame (600 ft slabs) is drawn next to the first. Hovering a band shows its height in ft MSL and as pressure altitude, the number of landings and departures, the temperature with its spread, flights and replies, and the wind with the spread between flights (Fig. 5c).



(a) Overlay, Analysis: Stability. **(b)** Zoom: an inversion at 2 600–3 000 ft MSL. **(c)** Hover on the 1 400 ft band: 13 flights, 127 replies.

Figure 5. Airport low-level profile from 19 landings, 9 October 2026, 11–12 UTC. The area profile (grey, 1 000 ft layers) shows no inversion; the airport profile resolves a 1.2 K inversion below 3 000 ft MSL.

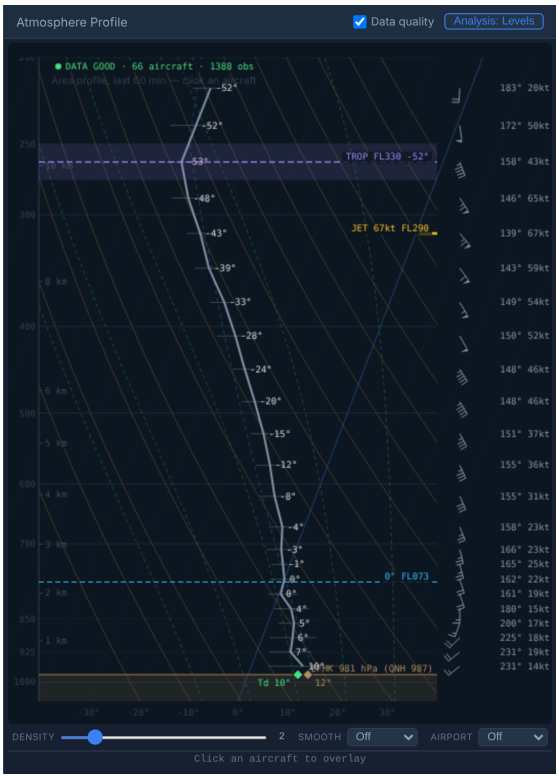
How far to trust it. A single reply on approach scatters by 3–4 K, so a band needs several flights; check the number in the hover. At night and in the early morning there are few flights, and the profile can have gaps. Departures climb 4–5 times faster than an approach descends, so they give fewer replies per band, and they turn earlier, so fewer winds (temperatures are not affected by turns); turboprops report no Mach number and give winds only. In the first weeks the approach temperatures averaged about 0.4 K colder than the area profile at the same pressure — a mix of real differences (the area profile includes aircraft over the sea and inland) and of calibration; compare shapes rather than single tenths.

10 The Analysis button

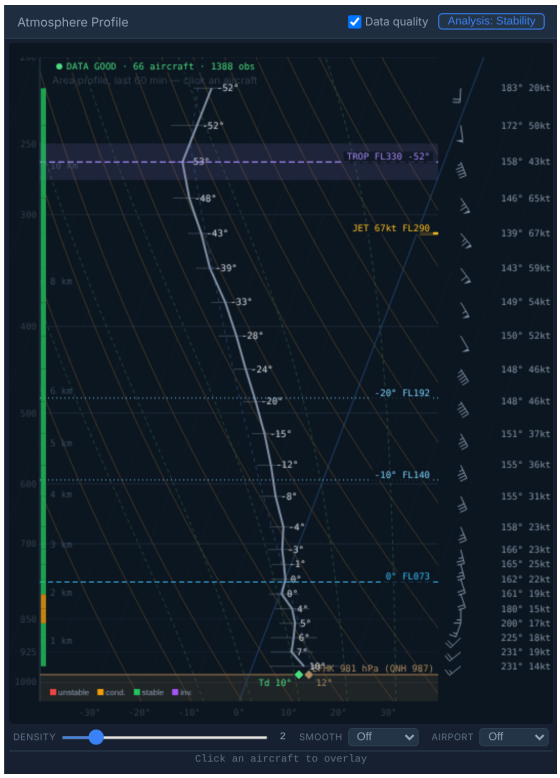
The Analysis button at the right of the panel title runs an automatic analysis of the area profile shown (on the Live Map the last 60 minutes; on the Sounding page the selected window, hour or flight). Each click (or a turn of the mouse wheel over the button) steps to the next level; each level adds to the previous ones. Lines and bands are drawn across the plot, labels at its right edge (spread so that they do not overlap), and marker bars along the right edge and next to the pressure axis.

Table 1. Analysis levels.

Level	Adds
Off	nothing
Levels	tropopause, 0 °C level(s), maximum wind / jet stream, low-level jet
Stability	−10 and −20 °C levels, inversions, stability strip
Turbulence	wind shear and turbulence risk (Richardson number)
Advection	warm and cold air advection, contrail levels
Full	daytime thermal top; with the METAR dewpoint the surface parcel: LCL / cumulus base, CAPE, CIN, Lifted Index, equilibrium level, fog / low stratus flag



(a) Levels: tropopause band (violet) `TROP FL330 −52°`, jet `JET 67kt FL290` (yellow tick at the right edge), freezing level `0° FL073` (dashed blue).



(b) Stability: adds the −10 / −20 °C levels (dotted) and the stability strip next to the pressure axis (key at the bottom left).

Figure 6. Analysis levels 1 and 2, 9 October 2026, 11–12 UTC.

10.1 Levels

Tropopause (violet band ±1 000 ft and dashed line, e.g. `TROP FL330 −52°`). Found with the WMO definition: the lowest level above 5 km where the lapse rate falls to 2 K/km or less and stays so, on average, over the next 2 km. A `?` after the label means there were few data above it, so the level is uncertain. At night, with little high-level traffic, it may not be found at all.

0 °C level (dashed blue, e.g. `0° FL073`). Every crossing of 0 °C is marked; a second or third crossing appears only when a warm or cold layer really differs by at least 1 K from 0 °C — small wiggles around zero are ignored.

Max wind / jet (yellow tick at the right edge): the strongest layer wind, labelled `JET` when it is 60 kt or more (the WMO jet-stream criterion), otherwise `MAX`.

Low-level jet (`LLJ`): a wind maximum below 5 000 ft of at least 25 kt that is at least 10 kt stronger than the wind within 4 000 ft above it. Typical at night over a cooled surface; gives low-level wind shear on approach.

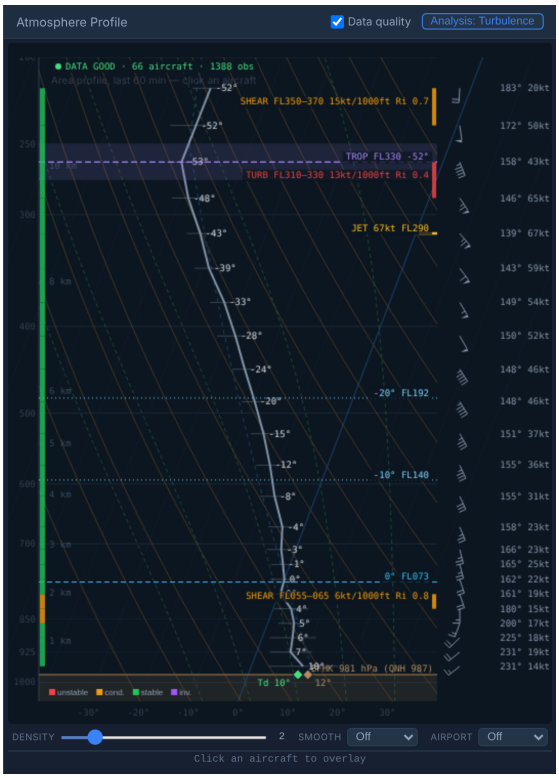
10.2 Stability

−10 and −20 °C levels (dotted). Together with 0 °C they bound the temperature range where airframe icing occurs *in cloud*. Whether there is cloud cannot be told from these data.

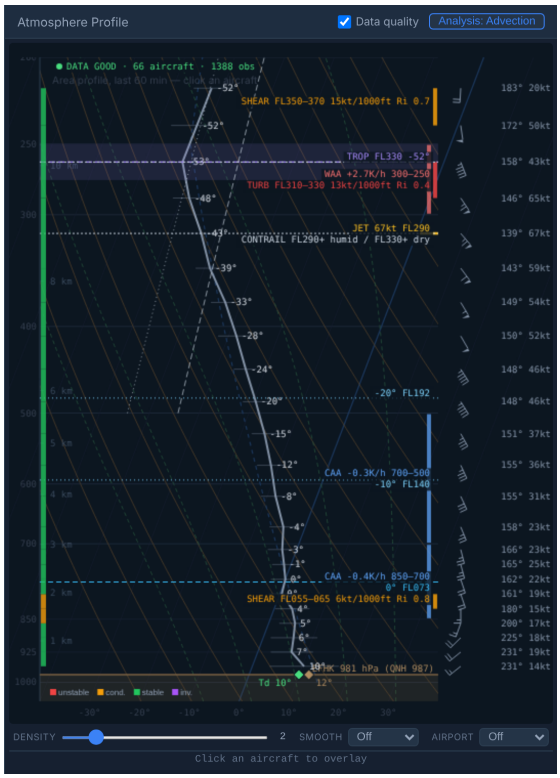
Inversions (orange shading, e.g. `INV +2.5° FL035–045`): layers where the temperature rises with height by more than twice the uncertainty of the two layer means (at least 1 K). Only below the tropopause — the stratosphere above is an inversion anyway (Fig. 8b).

Stability strip (narrow coloured bar right of the pressure axis): each layer's lapse rate compared with the adiabats — red absolutely unstable (> 9.8 K/km), amber conditionally unstable (between the moist and the dry rate), green stable, violet inversion. Above the tropopause it is usually violet.

Airport inversions and strip. With the Airport profile switched on, its significant inversions are shaded in amber and its own stability strip (600 ft slabs, amber frame) is drawn next to the first one (Section 9).



(a) Turbulence: `TURB FL310–330` (red bar) at the tropopause below the jet, `SHEAR` (amber) at FL350–370 and FL055–065.



(b) Advection: cold advection `CAA` (blue bar) at 850–500 hPa, warm `WAA` at 300–250 hPa; contrail threshold curves (dashed / dotted) and `CONTRAIL FL290+ humid / FL330+ dry`.

Figure 7. Analysis levels 3 and 4, 9 October 2026, 11–12 UTC.

10.3 Turbulence

For each pair of adjacent wind layers the **vertical wind shear** (vector difference, in kt per 1 000 ft) and the bulk **Richardson number**

$$Ri = \frac{N^2}{(\Delta V/\Delta z)^2}, \qquad N^2 = \frac{g}{\bar{\theta}}\,\frac{\Delta\theta}{\Delta z}$$

are computed, where θ is the potential temperature. Ri compares the stability, which damps vertical motion, with the shear, which drives it: below about 0.25 the flow tends to overturn into turbulence; up to 1 it is marginal.

- `TURB` (red bar and label): $Ri < 0.5$ and at least 4 kt per 1 000 ft;
- `SHEAR` (amber): $Ri < 1$ with at least 4 kt per 1 000 ft, or shear of at least 8 kt per 1 000 ft.

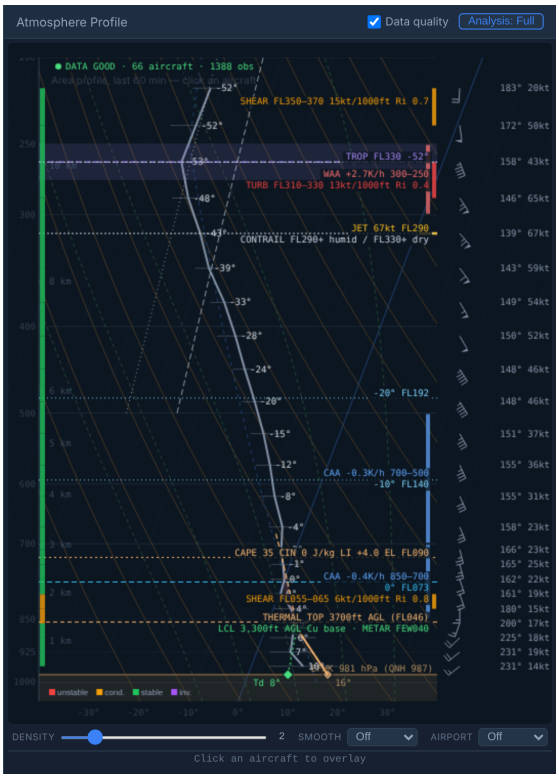
The shear must exceed twice the sampling uncertainty of the two mean winds, so random scatter alone does not trigger a flag. Typical places are just below and above the jet core, at the tropopause and in the lowest kilometre. Because the layers are 1 000–2 000 ft thick, thin shear layers are averaged out and real turbulence can be stronger and more local — treat the marks as indications, not as a turbulence forecast.

10.4 Advection and contrails

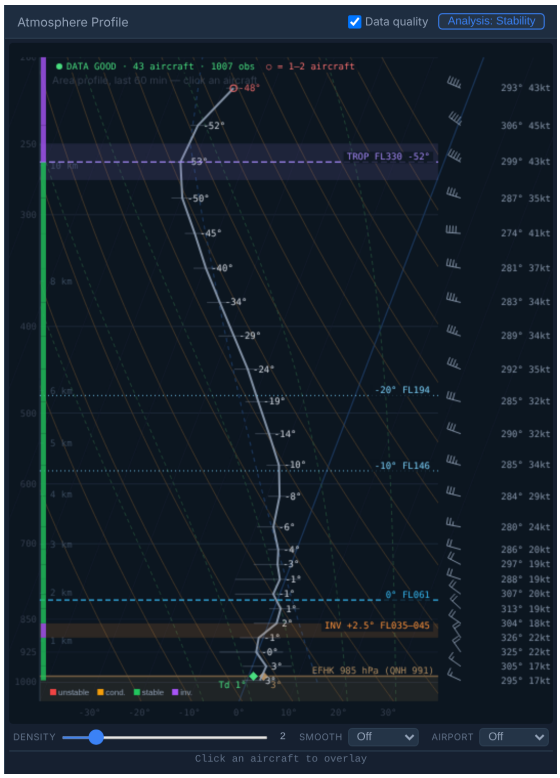
Warm and cold air advection (red / blue bar next to the shear bar, e.g. `CAA 0.4K/h 850–700`). In large-scale flow the change of wind with height (the *thermal wind*) is parallel to the isotherms of the layer. Wind turning clockwise with height (veering) therefore means that warm air is being carried in, anticlockwise (backing) cold air. The rate in K per hour is estimated for the layers 850–700, 700–500, 500–400, 400–300 and 300–250 hPa. The layer below 850 hPa is left out because there friction turns the wind by itself. A layer is shown only when both winds are at least 10 kt, the wind turns at least 10° and the change exceeds its uncertainty. Expect the profile to warm in WAA layers and to cool in CAA layers over the next hours; in the archive the signs agreed with the observed temperature changes on 8–10 October 2026. Rates above about 1–2 K/h near the jet are usually exaggerated by wind noise.

Contrail levels (`CONTRAIL FL290+ humid / FL330+ dry`). The Schmidt–Appleman criterion gives the temperature below which an engine exhaust plume becomes saturated and forms a contrail. It depends on pressure and on the humidity of the ambient air. Two threshold curves are drawn above 500 hPa: dashed for air saturated with respect to water (about −42 °C at 250 hPa), dotted for completely dry air (about −51 °C). Where the profile lies left of the dashed

curve, contrails form *if* the air is humid enough; left of the dotted curve they form at any humidity. Whether a contrail *persists* and spreads needs ice-supersaturated air, which cannot be seen without humidity data.



(a) Full with an illustrative METAR 16/08, FEW040: `THERMAL TOP 3700ft AGL (FL046)`; the parcel (dashed) has its `LCL 3,300ft AGL Cu base` next to `METAR FEW040`, CAPE 35 J/kg, Lifted Index +4.0, `EL FL090`: shallow fair-weather cumulus.



(b) 10 October 2026, 04–05 UTC, Stability: after the low, cold air near the ground with an inversion `INV +2.5° FL035-045`, freezing level FL061, tropopause FL330.

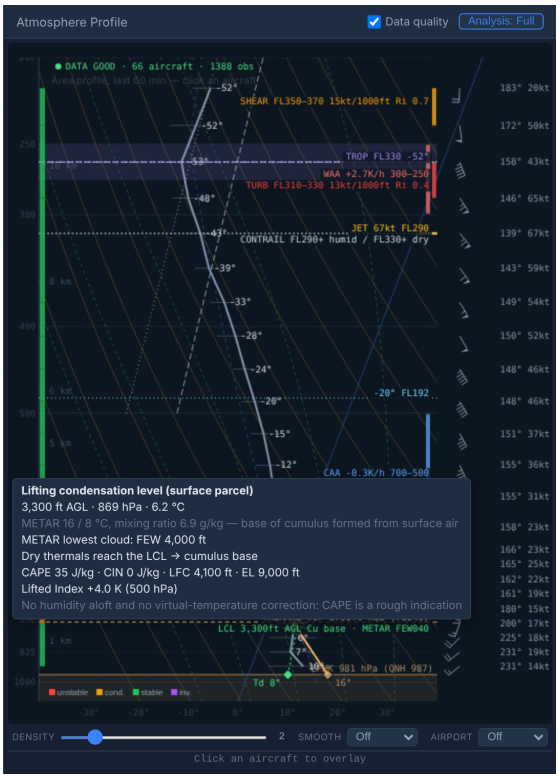
Figure 8. Thermal top with the surface parcel, and an inversion case.

10.5 Full: thermal top and surface parcel

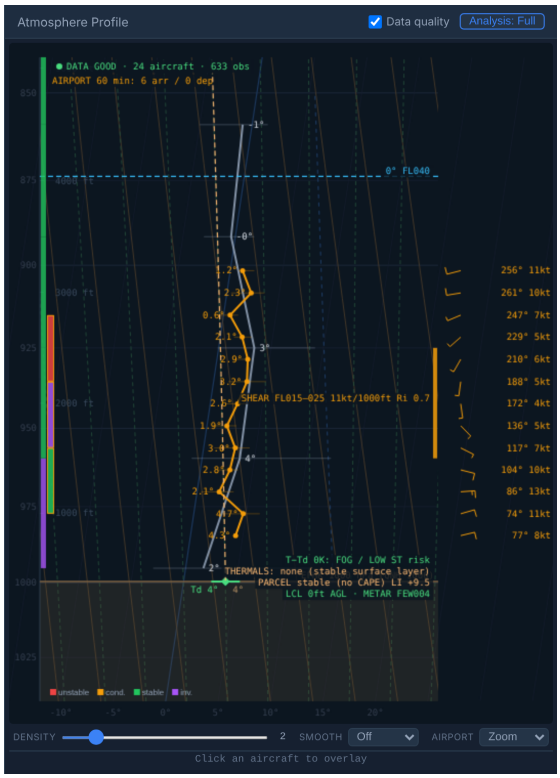
Dry thermal top. The surface temperature of the METAR, at the station pressure, is lifted along the dry adiabat (orange line) until it is no longer warmer than the measured profile (Fig. 8a). That height is the top of dry thermals — the depth of the daytime mixing layer, important for glider pilots, smoke and pollutant dispersion and low-level turbulence. It is labelled e.g. `THERMAL TOP 3700ft AGL (FL046)`. When the air just above the ground is already warmer than the surface air — at night and in the morning, or in cold cloudy weather — it shows `THERMALS: none (stable surface layer)`. When it cannot be computed it shows `THERMAL TOP: n/a` with the reason (no METAR temperature, too few low-level data). No extra surface heating is added, so on sunny afternoons the real top is often a few hundred feet higher.

Surface parcel with the METAR dewpoint (since 11 October 2026). The same surface air is followed further, now with its moisture:

- a green dotted **mixing-ratio line** rises from the dewpoint diamond; where it meets the dry adiabat from the temperature is the **LCL**, marked with a short green bar and labelled e.g. `LCL 3,300ft AGL Cu base · METAR FEW040`. `Cu base` is added when the dry thermals reach the LCL, i.e. surface heating can actually form cumulus there. The lowest METAR cloud group is printed beside it for comparison;
- above the LCL the parcel follows the **moist adiabat** (orange dashed). Where it is warmer than the profile, the area between them is shaded light red — that area is the **CAPE**. The label gives `CAPE ... CIN ... J/kg`, the **Lifted Index** and the **equilibrium level** `EL FL...`, where the parcel becomes colder again: roughly the top of a convective cloud. Without positive area it reads `PARCEL stable (no CAPE) LI +9.2`;
- a spread of 1.5 K or less adds `T−Td 0K: FOG / LOW ST risk`;
- hovering the LCL bar lists all values (Fig. 9a).



(a) Hover on the LCL bar of Fig. 8a: LCL 3 300 ft AGL, 869 hPa, 6.2 °C; mixing ratio 6.9 g/kg; CAPE, CIN, LFC, EL and Lifted Index.



(b) 11 October 2026, 06–07 UTC, real METAR 06:50 UTC 04/04, FEW004 BKN006, Q1006; Zoom, Full, airport profile from 6 landings.

Figure 9. The surface parcel: details on hover, and a low stratus morning.

Reading it — two cases. In Fig. 8a (illustrative) the surface air at 16/08 °C rises dry to 3 300 ft, where it condenses; the thermals reach 3 700 ft, so cumulus form with their bases near the LCL — matching the `FEW040` of the METAR. Above, the parcel is only slightly warmer than the profile up to about FL090 (CAPE 35 J/kg): small, flat cumulus, no showers. In Fig. 9b (real data) temperature and dewpoint are both 4 °C: the LCL is at the ground, the flag reads `FOG / LOW ST risk`, there are no thermals and the parcel is stable (LI +9.5). The airport profile shows a nearly isothermal, moist layer of 2–4 °C from 600 to 3 000 ft with the wind veering from 077° below to 256° above (`SHEAR FL015-025`): a stratus deck at 600 ft under a turning wind. Here the LCL does not predict the cloud base — stratus forms by cooling and mixing of a moist layer, not by thermals — it simply says that the surface air is saturated; the METAR `BKN006` is the cloud information.

Limits of the parcel. Only the surface dewpoint is known; the humidity above is not, so the parcel is assumed to keep its moisture (well-mixed boundary layer, typical on sunny afternoons). Parcel and environment use the plain temperature, without the virtual-temperature correction for water vapour, so CAPE is a rough indication (it tends to be underestimated by some tens of J/kg). At night the surface parcel is usually stable and the LCL is mainly a check against low cloud. The METAR is refreshed every 10 minutes; after a restart of the server, reload the page once the first METAR has arrived.

11 The Sounding page

The Sounding page draws the same diagram, sized to the window, next to a table of the layer values (Fig. 10). The buttons at the top left select the source:

- Area · now: the area profile of the last 30 minutes, 60 minutes, 2 or 3 hours, with the current METAR (ground line, dewpoint, parcel);
- Area · hour: the hourly profile archive, one stored profile per completed UTC hour, kept after the observations are purged. Since 11 October 2026 the METAR of that hour (temperature, dewpoint, QNH, lowest cloud) is stored with it, so the ground line and the surface parcel are shown for past hours too;
- Flight: the profile of one climbing or descending flight (also opened from the Flights page). No METAR is drawn.

The Analysis, Smooth, Airport and Data quality controls work as on the Live Map and share their settings with it. The line above the table gives the observations and aircraft used; the table's A/C column gives the number of different aircraft in each layer, red for one or two. Stored hours from 11 October 2026 on carry the aircraft count; older hours and single flights have no badge. The airport low-level profile follows the shown period: the selected window, that hour, or the

flight's time ±15 minutes (its landing or take-off). The aircraft overlay and the Density slider belong to the Live Map only.

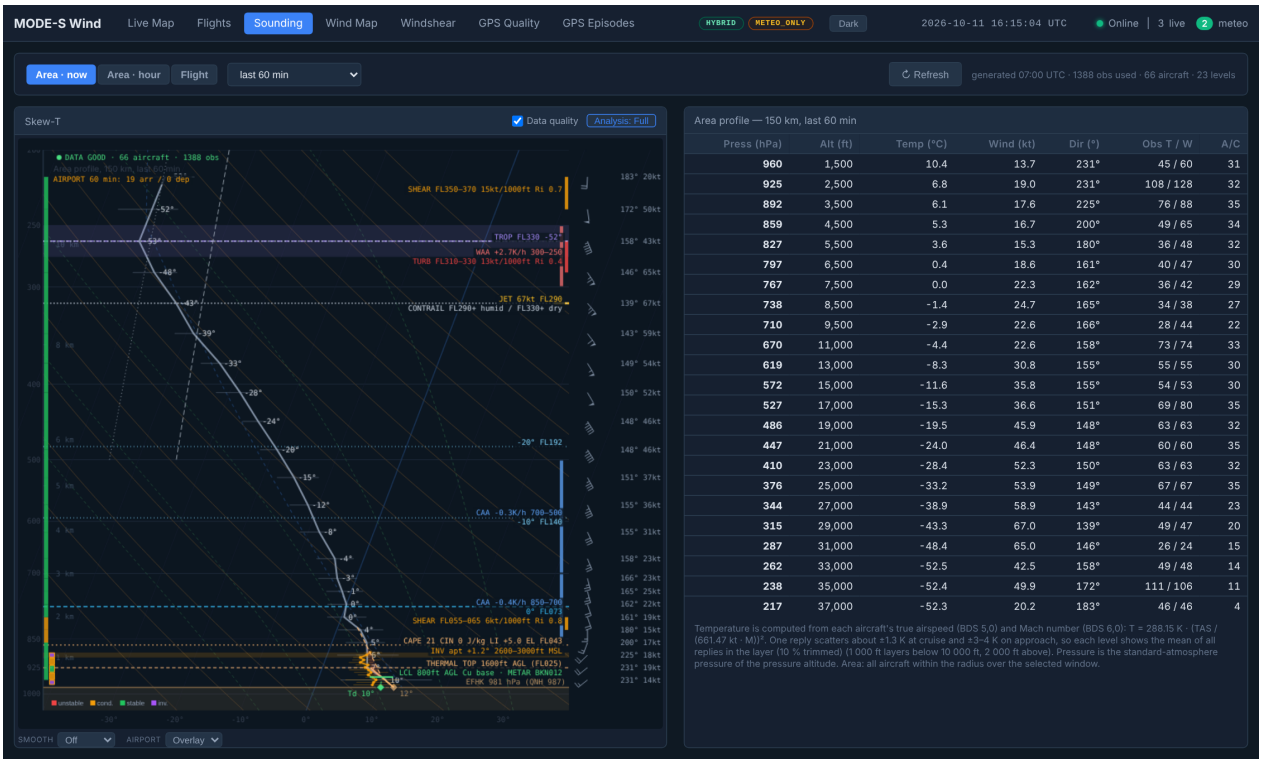


Figure 10. The Sounding page, Area · now, with the 9 October 2026 11–12 UTC profile (66 aircraft; A/C column at the right of the table), Airport: Overlay and Analysis: Full (illustrative METAR 12/10, BKN012: `LCL 800ft AGL Cu base · METAR BKN012`).

12 Comparison profiles: radiosonde, MEPS model and RH strip

Three tick boxes in the panel title — on the Sounding page above the diagram — draw other profiles next to the aircraft (Fig. 11). They are off by default; the choices are remembered in the browser and shared between the Live Map and the Sounding page.

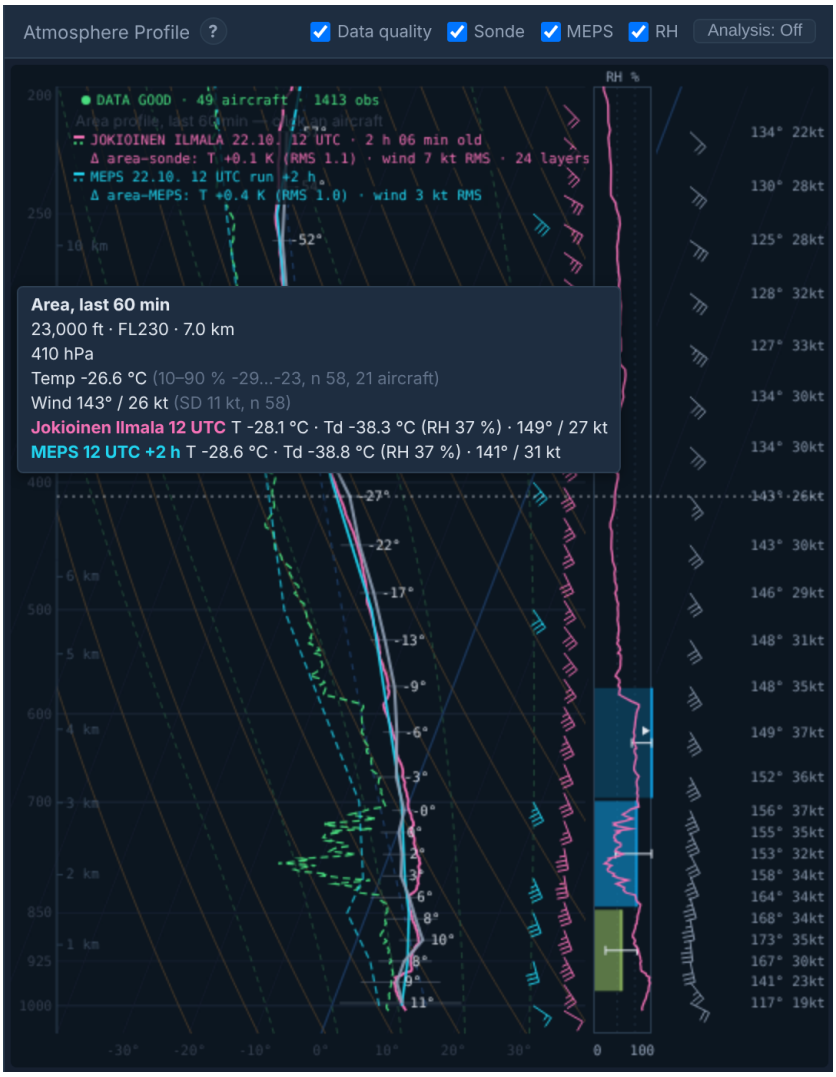


Figure 11. Live Map, 22 October 2026 about 13:40 UTC, with Sonde, MEPS and RH on. At 410 hPa the aircraft give −26.6 °C and 143°/26 kt, the 12 UTC Jokioinen sounding −28.1 °C and 149°/27 kt, the MEPS model over the same area −28.6 °C and 141°/31 kt. The labels at the top left give the mean differences: area − sonde +0.1 K (RMS 1.1) and 7 kt, area − model +0.4 K (RMS 1.0) and 3 kt. Right of the diagram the RH strip: the aircraft moisture indicator as bars for three layers and the sonde's relative humidity as a pink line.

Sonde — the Jokioinen radiosonde. FMI launches a balloon at Jokioinen (WMO 02963, about 100 km north-west, inland) at about 23:30, 05:30 and 11:30 UTC for the 00, 06 and 12 UTC soundings; MODE-S Wind fetches it from FMI open data while it is still climbing. Its temperature is drawn as a pink line, its dewpoint as a green dashed line — humidity aloft, which the aircraft cannot give — and its winds as pink barbs inside the right edge of the plot. The label gives the station, the launch, the age of the sounding and, for partial soundings, how high it reaches, and on the second line the mean difference area − sonde (temperature bias and RMS, wind RMS and the number of layers compared). The lines fade when the sounding is more than 6 and 12 hours old. The Live Map shows the latest sounding; the Sounding page the newest one launched by the shown hour or flight, or any stored sounding picked by hand.

MEPS — the weather model. MEPS is the 2.5 km weather model that FMI runs together with the Norwegian and Swedish weather services. MODE-S Wind fetches its main run every 3 hours from MET Norway's open data archive and stores, for every hour, the model averaged over the same 150 km area as the aircraft profile. It is drawn in teal: temperature solid, dewpoint dashed, winds as teal barbs a little inside the sonde's. The label gives the run and the hours after its start (`MEPS 22.10. 12 UTC run +2 h`) and the mean difference area − model; the hover read-out adds the model's temperature, dewpoint, relative humidity and wind at the level. A run appears in the archive about 2–3 hours after its start, so on the Live Map the present hour is a short forecast, +3 to +5 h, marked `(forecast)`; on the Sounding page past hours and flights use the model of that time.

Why MEPS has fewer levels. The open MEPS files give the model only on standard pressure levels; in the range of the diagram these are 1000, 925, 850, 800, 700, 500, 400, 300, 250 and 200 hPa — one value every 300 to 2 500 m, against a layer every 1 000 or 2 000 ft in the aircraft profile. The largest gap, between 700 and 500 hPa, spans about 10 000 to

18 000 ft. The model itself computes 65 levels, but these are published separately and would need several times the data on every fetch. The model lines are therefore straight between the levels, there is one barb per level (close ones, such as 850 and 800 hPa, may merge into one in the small Live Map panel), and the hover interpolates between them. A thin inversion or a sharp tropopause that the aircraft show between two model levels is therefore missing from the model line: that is the spacing of the levels, not a disagreement, and not missing data — waiting for more fetches adds hours, not levels.

RH (trial) — the moisture strip. A narrow strip between the diagram and the wind barbs, 0–100 %. Its bars are the trial moisture indicator from the aircraft themselves — moist air is lighter, so the thickness of a layer measured by climbing and descending aircraft, minus the Mach/TAS temperature, tells its moisture — for 1 000–5 000, 5 000–10 000 and 10 000–15 000 ft: brown *dry* (index below 40 %), green *moist*, blue *humid* (70 % or more), with its ± range, faint when uncertain. With Sonde on, the radiosonde's relative humidity is drawn in the strip as a pink line. The indicator is a first trial and a tendency only, uncertain by about ±30–50 %; how it works is explained in the Sounding page guide.

Reading them together. The three answer different questions. The sonde is a precise point measurement, but 100 km away and up to 12 hours old. The model is there every hour over the same area, but it is a model, on 10 levels. The aircraft are the live measurement over the area. When the aircraft agree with both, trust the profile. When they agree with the model but not with the sonde, the difference is usually the distance or the age of the sounding — a front between Helsinki and Jokioinen, or the air mass has changed since the launch. When they disagree with both at one level, look at the number of aircraft in that layer (Section 5). The Sounding page's MEPS check measures these differences every hour.

13 Limitations

Humidity only at the ground. Aircraft transponders do not report humidity (the MRAR format can, but practically no aircraft send it). The only humidity value the analysis uses is the METAR dewpoint at the airport; the sonde and the model dewpoints are drawn for comparison only (Section 12). It gives the spread, a fog / low-cloud indication and the surface parcel (LCL, CAPE, Lifted Index) — valid when the boundary layer is well mixed. There is no dewpoint curve aloft, so cloud layers above the ground, whether icing actually occurs, contrail persistence and the refractivity of the air (radio ducting) cannot be derived. The moist adiabats are reference lines.

Vertical resolution. A radiosonde reports every few metres; the area profile has layers of 1 000 ft (300 m) up to 10 000 ft and 2 000 ft (600 m) above. Thin features — a 20–50 m surface inversion, a sharp tropopause, a thin shear layer or a radio duct — are smoothed out or missed. Table 2 compares the automatic tropopause and freezing level with the Jokioinen soundings.

Table 2. Automatic analysis of the area profile against the full-resolution Jokioinen radiosonde (WMO 02963, about 100 km north-west of the receiver), pressure altitude.

Sounding	Tropopause, sonde / area	0 °C level, sonde / area
8 Oct 2026 12 UTC	11.4 / 11.3 km	1.6 / 2.2 km
9 Oct 2026 00 UTC	11.1 km / not found	2.4 / 2.1 km
9 Oct 2026 12 UTC	10.5 / 10.1 km	2.3 / 2.2 km
10 Oct 2026 00 UTC	9.9 / 9.4 km	0.8 / 1.6 km

Sampling. The profile exists only where aircraft fly. Cruise levels are well sampled by day, low levels mainly along the EFHK approach and departure paths, and the night hours (about 22–02 UTC) may have only a few aircraft — the data badge shows LOW then, and single-aircraft layers are ringed (Section 5). Layers need at least 3 replies; check the number of replies and aircraft and the 10–90 % range in the hover read-out. The airport profile depends on the traffic at EFHK: at night it may come from one or two flights. The area is 150 km in radius, so the profile is an area average over an hour, not a point measurement at one moment — a front passing through the area blurs it.

Accuracy. Temperatures from Mach and TAS are good at cruise (about ±1 K for layer means) but scatter by 3–4 K per reply at approach speeds, which is why the range bars are wide low down (Section 3); low-level layer means are therefore less certain, especially with few aircraft. Winds depend on correct heading, TAS and the absence of turns; with the per-airframe heading correction they are good to a few knots at cruise and less reliable at low speed and in turbulence.

Heights. The vertical scale is pressure; the km markers and flight levels are pressure altitude. True heights differ with QNH and air temperature. The thermal top, the LCL and the parcel levels are given in feet above ground, computed hypsometrically; the airport profile is in feet above mean sea level.

Automatic analysis. The analysis uses simple, documented rules on averaged data. It can miss features, mark marginal ones, or be fooled by sparse data. Use it as a guide to the diagram, not as an official weather product; for flight planning use official forecasts and soundings.

14 Quick reference

Symbol	Meaning
grey line, values	area temperature profile, last 60 min, 150 km
coloured line, circle	selected aircraft and its latest point
thin horizontal bars	10–90 % range of the temperatures in a layer (wide low down: Mach / TAS steps)
`DATA GOOD / FAIR / LOW`	data badge: different aircraft and observations behind the profile
red ring, red value	layer measured by only 1–2 aircraft
dashed blue line	ISA standard atmosphere
bold blue diagonal	0 °C isotherm
faint brown solid lines	dry adiabats (every 10 K)
faint green dashed lines	moist (saturated) adiabats
brown line, shading, diamond	ground (station pressure from METAR QNH), surface temperature
green diamond `Td`	METAR dewpoint at the ground
amber line, dots, barbs	airport low-level profile (200 ft bands, ft MSL)
amber shading `INV apt`, amber-framed strip	airport inversion, airport stability (600 ft slabs)
violet band `TROP`	tropopause
dashed blue `0°` / dotted `−10°`, `−20°`	isotherm levels
yellow tick `JET` / `MAX` / `LLJ`	maximum wind, jet stream, low-level jet
orange shading `INV`	inversion
strip at the pressure axis	stability: red unstable, amber conditional, green stable, violet inversion
red / amber bar `TURB` / `SHEAR`	turbulence risk / strong shear
red / blue bar `WAA` / `CAA`	warm / cold air advection (K/h)
dashed / dotted curves `CONTRAIL`	contrail thresholds for humid / dry air
orange line `THERMAL TOP`	dry adiabat from the surface and the thermal top
green dotted line, green bar `LCL`	mixing-ratio line from the dewpoint, lifting condensation level (METAR cloud beside it)
orange dashed line, red shading	parcel curve (moist adiabat above the LCL), CAPE area
`CAPE / CIN / LI / EL`	convective energy, inhibition, Lifted Index, equilibrium level
`FOG / LOW ST risk`	spread T−Td ≤ 1.5 K
pink line, green dashed, pink barbs	Jokioinen radiosonde: temperature, dewpoint, winds (fade with age)
teal line, teal dashed, teal barbs	MEPS model over the area (10 levels); `(forecast)` : +3 to +5 h
brown / green / blue bars in the RH strip	aircraft moisture indicator (trial): dry / moist / humid; pink line: sonde RH

A Formulas

- Mach/TAS temperature: $T = 288.15\,(\mathrm{TAS}/(661.47\,M))^2$ K; used for Mach ≥ 0.2, TAS ≥ 60 kt and within 40 K of ISA.
- Pressure of pressure altitude h (m): $p = 1013.25\,(1 - 0.0065\,h/288.15)^{5.25588}$ hPa below 11 km.
- Station pressure: $p_{\mathrm{stn}} = \mathrm{QNH}\,(1 - 2.25577 \times 10^{-5}\,z)^{5.25588}$, z = airport elevation (m).
- Dry adiabat: $T = \theta\,(p/1000)^{R/c_p}$, $R/c_p = 0.2857$.
- Moist adiabat: $\dfrac{dT}{dp} = \dfrac{RT + Lr_s}{p\,(c_p + L^2 r_s \varepsilon/(RT^2))}$, integrated from 1000 hPa; r_s saturation mixing ratio, $L = 2.501 \times 10^6$ J/kg, $\varepsilon = 0.622$.
- Layer thickness (hypsometric): $\Delta z = (R\bar{T}/g)\ln(p_1/p_2)$.
- Thermal advection: $-\vec{V}\cdot\nabla T = -\dfrac{f}{R\ln(p_1/p_2)}\,(u_1 v_2 - v_1 u_2)$, $f = 2\Omega\sin\varphi$.
- Mixing ratio: $w = \varepsilon\,e/(p-e)$, $e = 6.112\exp\!\big(17.67\,T_d/(T_d + 243.5)\big)$ hPa (T_d in °C).
- LCL (Bolton 1980): $T_L = 1/\big(1/(T_d - 56) + \ln(T/T_d)/800\big) + 56$ (K), $p_L = p_0\,(T_L/T)^{c_p/R}$.
- CAPE $= R\int_{p_{EL}}^{p_{LFC}} (T_p - T_e)\,d\ln p$, CIN the same integral of the negative part between the ground and the LFC; Lifted Index $= T_e(500) - T_p(500)$.
- Schmidt–Appleman: $G = \dfrac{EI_{\mathrm{H_2O}}\,c_p\,p}{\varepsilon\,Q\,(1-\eta)}$ with $EI = 1.25$, $Q = 43.2$ MJ/kg, $\eta = 0.3$; $T_{LM} = -46.46 + 9.43\ln(G - 0.053) + 0.72\ln^2(G - 0.053)$ °C (saturated air), $T_{LC} = T_{LM} - e_s(T_{LM})/G$ (dry air).

Written with the assistance of Claude (Anthropic). Screenshots: MODE-S Wind, dark theme, with archived area and airport profiles of 9, 10 and 11 October 2026 (aircraft counts recomputed from the stored observations). The surface values in Figs. 1, 8a, 8b, 9a and 10 are illustrative METARs; Fig. 9b uses the real EFHK METAR of 11 October 2026, 06:50 UTC. Fig. 11: the database of 22 October 2026 13:37 UTC with the 12 UTC Jokioinen sounding (FMI open data, CC BY 4.0) and MEPS model data from MET Norway (thredds.met.no, CC BY 4.0).